

FLOODGAUGE · REQUIREMENTS

FloodGauge requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/03-requirements.md @ d4d6862		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from FLG-CAL-001; R16 restated to FloodGauge-specific parts under FLG-DDR-001 D1
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

FloodGauge requirements

These requirements are checked by calculation in FLG-CAL-001 v0.1. The design choices behind them are adopted for TRL 3 work under Amish's 2026-09-25 instruction, open for his review (FLG-DDR-001). One target changed at TRL 3: R16 now covers FloodGauge-specific parts only, with the FieldNode core costed in its own repo (D1). Two further changes to targets are proposed, awaiting Amish, and are not applied: R1 and R3 (see below). The status column states "not met" plainly.

Table 1. Requirements and status at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (FLG-CAL-001)
R1	Measure water depth on the street at the gauge point	0 to 600 mm above the road surface, head mounted 2.5 to 3.5 m above the road	Range calculation against sensor data	Met on paper: 2.95 to 2.35 m range, radar to 20 m. The 3.5 m upper limit may conflict with vehicle clearance (proposal below)
R2	Street depth accuracy	±10 mm over -20 to 50 °C air, still water	Error budget; later bench test over a tank	Met on paper with radar: ±6.3 mm root sum square (±11.3 mm worst-case sum), module accuracy assumed. Not met by the ultrasonic variant (±19.5 mm compensated)
R3	Measure water level in the catch basin	From 50 mm above the basin floor to the underside of the grate	Geometry of stilling tube and head	Not met: continuous from 50 mm above the floor to 330 mm below the road; the top 290 mm is covered by the wet probe only
R4	Drain level accuracy	±20 mm	Error budget	Met on paper: ±11.0 mm with the NTC in the head (±19.7 mm without)
R5	Sampling rate	Every 60 s normally; every 10 s when level is above the first alert band or rising faster than 20 mm/min	Sampling logic review	Met by design
R6	Alert latency	120 s or less from threshold crossing to message at the alert service, 95 % of events	Latency budget	Met on paper: 82 s at the 95th percentile on a private gateway (85 s on a city server); 110 s worst case without a lost packet
R7	Radio airtime within network rules	Within EU868 1 % duty cycle always; within The Things Network fair use (30 s uplink airtime per day) in normal mode	Airtime calculation	At risk: normal mode 23.7 s/day at SF9; event mode within 1 % at SF7 to SF10 but not at SF11 or SF12 unless reporting stretches to 99 s or 181 s; normal mode exceeds fair use at SF10 and slower
R8	Energy autonomy	5 days without sun at event-mode sampling	Power budget against FieldNode	Met on paper: 68 days nominal, 47 days at -20 °C; 9.3 % of the FieldNode 100 mW allowance
R9	Survive submersion of the drain head	IP68, 2 m, 72 h, stormwater with silt	Enclosure design review; later immersion test	At risk: electronics potted, but the transducer face keeps the module's IP67 seal
R10	Reject false readings	No alert caused by a vehicle, person or debris under the street head; target 1 false alert per gauge per year or fewer	Plausibility logic review; later field log	Not verifiable at TRL 3: rules defined, including a recorded dry-road background against the curb echo
R11	Privacy	Levels, status and battery only; no camera or microphone; no images or audio leave the device	Design review	Met by design
R12	Installation	Two-person crew, 90 min or less, from the surface; no road	Installation sequence review	Not met: 90 min for the surface work alone; the conduit needs coring and trenching

ID	Requirement	Target	Verification (TRL 3 or later)	Status (FLG-CAL-001)
		drilling; no entry into the catch basin		
R13	Level datum	Head height surveyed to the road surface at the gauge point within ± 5 mm; checked yearly against the depth marker	Survey method	Met on paper: tape survey plus a recorded dry-road radar reading
R14	Operating environment	-20 to 50 °C air; UV, rain and road spray; FieldNode charges only from 0 to 45 °C	Material and cell review	At risk: inherits FieldNode's interior heat finding (FND R2, R3); the drain head module is rated -15 to 60 °C
R15	Mounting	Fits 40 to 60 mm poles with band clamps; total added mass on the pole 5 kg or less	Mass estimate	Met on paper: 4.62 kg; clamp twist factor only 1.45 in a 35 m/s gust
R16	Cost	FloodGauge-specific parts \$150 or less at quantity 1; FieldNode costed separately (FND-CAL-001)	Priced BOM	Met on paper: \$145.50 (\$271.50 with the \$126.00 FieldNode core)
R17	Open data	Levels published in an open, documented format (JSON or CSV) through the gateway	Design review	Met by design

■ Assumptions

- The road surface at the gauge point is the datum for street depth. Depth equals the surveyed head height minus the measured range.
- "Event" means water above the first alert band or rising faster than 20 mm/min.
- FieldNode figures (usable energy, allowance, airtime per uplink, mass, cost, charge temperature window) come from FND-CAL-001 in the FieldNode repo.
- Alert depths of 150 mm and 300 mm follow NWS guidance on water that can knock over an adult and carry away a car ([NWS](#)). They are defaults under FLG-DDR-001 D6, to be set locally with the partner city.
- The remaining assumptions for every status above are in FLG-CAL-001, Table 1.

■ Requirements not met or at risk

- **R3 not met:** the drain head cannot sit higher than 300 mm below the road, and its 30 mm blind zone leaves 290 mm below the grate with only the wet probe. Proposed, awaiting Amish: restate R3 as "continuous reading from 50 mm above the basin floor to 330 mm below the road, plus a drain-full signal above that".
- **R12 not met:** the conduit from basin to pole is civil work. Proposed, awaiting Amish: a surface cable route along the curb face under a steel cover for pilots.
- **R7 at risk:** slow spreading factors need a longer event interval; event reporting belongs on a private or city gateway (D7).
- **R9 at risk:** the transducer face seal decides immersion survival.

- **R14 at risk:** interior heat in FieldNode, and the drain module's -15 °C lower rating.
- **R1 note:** at 2.95 m the head is inside a 4.0 to 4.3 m vehicle envelope at the curb. Proposed, awaiting Amish: widen the mounting band to 2.5 to 5.0 m and set the height to local clearance rules.
- **R2** would not be met if the ultrasonic street variant were used.